

# Math 1352

## 2006 Spring Semester

| Monday                                              | Tuesday                                                                                                                                                                                                            | Wednesday                                                   | Thursday                                                                                                                                                                                                                                     | Friday             |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <i>January 9</i>                                    | <i>January 10</i>                                                                                                                                                                                                  | <i>January 11</i>                                           | <i>January 12</i><br>6.1: area between curves, area using vertical strips, area using horizontal strips                                                                                                                                      | <i>January 13</i>  |
| <i>January 16</i><br><i>Martin Luther King, Jr.</i> | <i>January 17</i><br>6.2: method of cross sections, method of disks and washers, method of cylindrical shells<br><br><div>HW 01 Due: 6.1</div>                                                                     | <i>January 18</i><br><br><br><div>WW 01 Due: 6.1</div>      | <i>January 19</i><br>6.3: polar coordinates, polar graphs, intersection of polar curves, area for polar coordinates                                                                                                                          | <i>January 20</i>  |
| <i>January 23</i>                                   | <i>January 24</i><br>6.4: arc length of a curve, area of a surface of rotation, polar arc length and surface area<br><br><div>HW 02 Due: 6.2-6.3</div>                                                             | <i>January 25</i><br><br><br><div>WW 02 Due: 6.2-6.3</div>  | <i>January 26</i><br>6.5: work, fluid pressure and force, centroids and moments in the plane, theorem of Pappus                                                                                                                              | <i>January 27</i>  |
| <i>January 30</i>                                   | <i>January 31</i><br>7.1-7.2: review of substitution, use of tables, (begin) integration by parts<br><br><div>HW 03 Due: 6.4-6.5</div>                                                                             | <i>February 1</i><br><br><br><div>WW 03 Due: 6.4-6.5</div>  | <i>February 2</i><br>7.2-7.3: integration by parts, repeated integration by parts, definite integration by parts, (begin) trigonometric substitution: powers of sine and cosine<br><i>Groundhog Day</i><br><div>Review Session 7:00 pm</div> | <i>February 3</i>  |
| <i>February 6</i>                                   | <i>February 7</i><br><b>Exam I</b><br>(6.1-6.5)                                                                                                                                                                    | <i>February 8</i>                                           | <i>February 9</i><br>7.3: trigonometric substitution: powers of secant and tangent, quadratic-form integrals                                                                                                                                 | <i>February 10</i> |
| <i>February 13</i>                                  | <i>February 14</i><br>7.4: partial fraction decompositions, Heaviside method, integrating rational functions, rational functions of sine and cosine<br><i>Valentine's Day</i><br><br><div>HW 04 Due: 7.1-7.3</div> | <i>February 15</i><br><br><br><div>WW 04 Due: 7.1-7.3</div> | <i>February 16</i><br>7.5: strategies for integration                                                                                                                                                                                        | <i>February 17</i> |

# Math 1352

## 2006 Spring Semester

| Monday                                       | Tuesday                                                                                                                                                        | Wednesday                                                                    | Thursday                                                                                                                                        | Friday                                  |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <i>February 20</i><br><i>President's Day</i> | <i>February 21</i><br>7.6: first-order linear differential equations, applications of first-order equations<br><br>HW 05 Due: 7.4-7.5                          | <i>February 22</i><br><i>Washington's Birthday</i><br><br>WW 05 Due: 7.4-7.5 | <i>February 23</i><br>7.7: improper integrals with infinite limits of integration, improper integrals with unbounded integrands                 | <i>February 24</i>                      |
| <i>February 27</i>                           | <i>February 28</i><br>7.8: hyperbolic functions, derivatives and integrals of hyperbolic functions, inverse hyperbolic functions<br><br>HW 06 Due: 7.6-7.7     | <i>March 1</i><br><br>WW 06 Due: 7.6-7.7                                     | <i>March 2</i><br>8.1: sequences, limits of sequences, bounded sequences, monotonic sequences, sandwich (squeeze) theorem for sequences         | <i>March 3</i>                          |
| <i>March 6</i>                               | <i>March 7</i><br>8.2: definition of infinite series, general properties, geometric series, applications of the geometric series<br><br>Review Session 7:00 pm | <i>March 8</i>                                                               | <i>March 9</i><br><b>Exam II</b><br>(7.1-7.7)                                                                                                   | <i>March 10</i>                         |
| <i>March 13</i>                              | <i>March 14</i>                                                                                                                                                | <i>March 15</i>                                                              | <i>March 16</i>                                                                                                                                 | <i>March 17</i><br><i>St. Patrick's</i> |
| Spring Break                                 |                                                                                                                                                                |                                                                              |                                                                                                                                                 |                                         |
| <i>March 20</i>                              | <i>March 21</i><br>8.3: divergence test, series of non-negative numbers, integral test, p-series<br><br>HW 07 Due: 7.8-8.2                                     | <i>March 22</i><br>Last Day to Drop a Course<br><br>WW 07 Due: 7.8-8.2       | <i>March 23</i><br>8.4: direct comparison test, limit comparison test                                                                           | <i>March 24</i>                         |
| <i>March 27</i>                              | <i>March 28</i><br>8.5: 8.4-8.5: ratio test, root test<br><br>HW 08 Due: 8.3-8.4                                                                               | <i>March 29</i><br><br>WW 08 Due: 8.3-8.4                                    | <i>March 30</i><br>8.6: alternating series test, error estimates for alternating series, absolute and conditional convergence, summary of tests | <i>March 31</i>                         |

# Math 1352

## 2006 Spring Semester

| Monday                                                     | Tuesday                                                                                                                                                                                                                                     | Wednesday                                                                 | Thursday                                                                                                                                          | Friday                               |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <i>April 3</i>                                             | <i>April 4</i><br>8.7: convergence of a power series, term-by-term differentiation and integration of a power series<br><br><div>HW 09 Due: 8.5-8.6</div>                                                                                   | <i>April 5</i><br><br><br><div>WW 09 Due: 8.5-8.6</div>                   | <i>April 6</i><br>8.8: Taylor and Maclaurin polynomials, Taylor's theorem, Taylor and Maclaurin series, operations on Taylor and Maclaurin series | <i>April 7</i>                       |
| <i>April 10</i>                                            | <i>April 11</i><br>9.1: introduction to vectors, vectors in component form, standard representations in the plane<br><br><div>HW 10 Due: 8.7-8.8</div>                                                                                      | <i>April 12</i><br><br><br><div>WW 10 Due: 8.7-8.8</div>                  | <i>April 13</i><br>9.2: coordinates in space, graphs in space, spheres and cylinders, vectors in space<br><br><div>Review Session 7:00 pm</div>   | <i>April 14</i>                      |
| <i>April 17</i>                                            | <i>April 18</i><br><b>Exam III</b><br>(7.8-8.8)                                                                                                                                                                                             | <i>April 19</i>                                                           | <i>April 20</i><br>9.3: definition and basic properties of dot product, angle between vectors, direction cosines, projections                     | <i>April 21</i>                      |
| <i>April 24</i>                                            | <i>April 25</i><br>9.4: definition and basic properties of the cross product, geometric interpretation of the cross product, applications of the cross product: area and torque, triple scalar product<br><br><div>HW 11 Due: 9.1-9.3</div> | <i>April 26</i><br><br><br><div>WW 11 Due: 9.1-9.3</div>                  | <i>April 27</i><br>9.5: parametric equations, parametrizing a curve, lines in space                                                               | <i>April 28</i>                      |
| <i>May 1</i>                                               | <i>May 2</i><br>9.6: forms for the equation of a plane in space, vector methods for measuring distances in space<br><br><div>HW 12 Due: 9.4-9.5</div>                                                                                       | <i>May 3</i><br>Individual Study Day<br><div>Review Session 3:00 pm</div> | <i>May 4</i>                                                                                                                                      | <i>May 5</i><br><i>Cinco de Mayo</i> |
| Period of No Exams                                         |                                                                                                                                                                                                                                             | WW 12 Due: 9.4-9.5                                                        | Period of No Exams                                                                                                                                |                                      |
| <i>May 8</i><br><div>Departmental Final Exam (10:30)</div> | <i>May 9</i>                                                                                                                                                                                                                                | <i>May 10</i>                                                             | <i>May 11</i>                                                                                                                                     | <i>May 12</i>                        |